1 Orthogonal Complements

Definition 1.1 (Orthogonal complement)**.** If U is a subset of V, then the *orthogonal complement* of U, denoted by $U^\perp$, is the set of all vectors in V that are orthogonal to every vector in U:

$$U^\perp := \{v \in V : (u, v) = 0 \text{ for every } u \in U\}.$$

Exercise 1.2. Prove the following:

(a) If U is a plane in $\mathbb{R}^3$ containing the origin, then $U^\perp$ is the line containing the origin that is perpendicular to U.

(b) If U is a line in $\mathbb{R}^3$ containing the origin, then $U^\perp$ is the plane containing the origin that is perpendicular to U.

Exercise 1.3. Prove that if $\{e_1, \ldots, e_m, f_1, \ldots, f_n\}$ is an orthonormal basis of V, then

$$(\text{span}(e_1, \ldots, e_m))^\perp = \text{span}(f_1, \ldots, f_n).$$

Theorem 1.4 (Properties of the orthogonal complement)**.**

(a) If U is a subset of V, then $U^{\perp}$ is a subspace of V.

(b) $\{0\}^{\perp} = V$.

(c) $V^{\perp} = \{0\}$.

(d) If U is a subset of V, then $U \cap U^{\perp} \subseteq \{0\}$.

(e) If G and H are subsets of V and $G \subseteq H$, then $H^{\perp} \subseteq G^{\perp}$.

Exercise 1.5. Prove Theorem 1.4.

Proposition 1.6. *Suppose U is a finite-dimensional subspace of V. Then $V = U \oplus U^{\perp}$. (*Hint*: Given an orthonormal basis $\{e_1, \ldots, e_n\}$ of V, we have seen before that we can write $v = u + w$, with $(u, w) = 0$, by taking $u = \sum_{i=1}^{n} (v, e_i) e_i$ and $w = v - u$.)*

Exercise 1.7. Prove Proposition 1.6.

Corollary 1.8 (Dimension of orthogonal complement)**.** *Suppose V is finite-dimensional and U is a subspace of V. Then*

$$\dim U^{\perp} = \dim V - \dim U.$$

Exercise 1.9. Prove Corollary 1.8.

Proposition 1.10. *Suppose U is a finite-dimensional subspace of V. THen*

$$(U^{\perp})^{\perp} = U$$

Exercise 1.11. Prove Proposition 1.10.

Suppose U is a subspace of V and we want to show that $U = V$. In some situations, the easiest way to do this is to show that the only vector orthogonal to U is 0, and then use the result below.

Corollary 1.12. *Suppose U is a finite-dimensional subspace of V. Then*

$$U^{\perp} = \{0\} \iff U = V.$$

Exercise 1.13. Prove Corollary 1.12.

Exercise 1.14. Suppose $\{e_1, \ldots, e_n\}$ is a set of vectors in V with $||e_k|| = 1$ for each $k = 1, \ldots, n$ and

$$||v||^2 = \sum_{i=1}^{n} |(v, e_i)|^2$$

for all $v \in V$. Prove that $\{e_1, \ldots, e_n\}$ is an orthonormal basis of V. (This provides a converse of a previous result.)

Exercise 1.15. Suppose $V = \mathscr{C}([-1,1])$ is the vector space of continuous real-valued functions on the interval $[-1,1]$ with inner product given by

$$(f,g) = \int_{-1}^{1} f(x)g(x)dx.$$

Let U be the subspace of V defined by

$$U = \{f \in \mathscr{C}([-1,1]) : f(0) = 0\}$$

(a) Show that $U^{\perp} = \{0\}$.

(b) Show that Propositions 1.6 and 1.10 do not hold without the finite-dimensional hypothesis.